DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech. **Branch: Elect. & Instrumentation** **Semester :VIIIth**

Subject Code & Name: BTEIEP801:4 DC POWER TRANSMISSION SYSTEMS

Max Marks: 60 **Date:07/07/2022** **Duration: 3.45 Hr.**

Instructions to the Students:

1. *All the questions are compulsory.*
2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.*
3. *Use of non-programmable scientific calculators is allowed.*
4. *Assume suitable data wherever necessary and mention it clearly.*

		(Level/CO)	Marks
Q. 1	**Solve Any Six of the following.**		**12**
A)	What are the requirements for the choice of converter configuration?		
B)	Define pulse number?		
C)	What is valve utilization factor?		
D)	State TUF meaning		
E)	Why the necessity of control in a DC link?		
F)	Define current margin		
G)	Draw the circuit of Graetz circuit.		
Q.2	**Solve Any Three of the following.**		**12**
A)	Write the advantages and disadvantages of HVDC Transmission Advantages		
B)	Explain **Thyristors-based LCCs** Technology.		
C)	State differences between LCC and CCC		
D)	Explain Commutation margin angle of 6 pulse LCC		
Q. 3	**Solve Any One of the following.**		**6**
A)	Draw and Explain schematic diagram of a12 pulse LCC circuit.		
B)	Explain Link commutated Converter.		
Q.4	**Solve Any Two of the following.**		**20**
A)	Draw and Explain HVDC Converter Station.		
B)	State and Explain different types of HVDC links.		
C)	Explain Converter control characteristics.		
Q. 5	**Write a short note on the following.**		**10**
A)	Double tuned and damped filters.		
B)	Reactive power requirement		

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